

Population Genetics

Frequency Dependent Selection

(a) Definition

Frequency dependent selection occurs when the fitness of a genotype depends on its frequency in the population. In other words, whether an allele is advantageous or disadvantageous changes depending on how common or rare it is.

This is different from the models we have studied previously, where fitness values (w_{11} , w_{12} , w_{22}) were constant regardless of allele frequency.

(b) Types of Frequency Dependent Selection

(i) Positive Frequency Dependent Selection

In positive frequency dependent selection, the fitness of a genotype increases as it becomes more common. Common types have an advantage simply because they are common.

Consequence:

- The common allele becomes even more common
- Rare alleles are at a disadvantage
- Eventually, one allele may go to fixation, and genetic diversity is lost
- However, if the population reaches fixation, there is no further change unless new variation arises

Examples:

- **Warning coloration in toxic species:** Predators learn to avoid a color pattern more quickly if it is common. A rare color pattern is not recognized and is attacked more often. Therefore, the common color pattern has higher fitness because predators have learned to avoid it.
- **Scaly-leg mite resistance in pigeons:** In some bird species, the allele conferring resistance to scaly-leg mites becomes more beneficial as it becomes more common because it reduces the overall parasite load in the population, benefiting all resistant individuals.

(ii) Negative Frequency Dependent Selection

In negative frequency dependent selection, the fitness of a genotype increases as it becomes rarer. Rare types have an advantage simply because they are rare.

Consequence:

- Rare alleles increase in frequency
- Common alleles are at a disadvantage
- This creates a balancing effect that maintains genetic diversity in the population
- Allele frequencies tend to oscillate or settle at an equilibrium where both alleles are maintained

Examples:

- **Rare male advantage:** In many species (e.g., fruit flies, chickens, humans), females show a preference for rare or novel male phenotypes. This gives rare male types a mating advantage, preventing any single male type from becoming too common. This is a classic example of negative frequency dependent selection.
- **Mimicry (Batesian mimicry):** A harmless species (mimic) evolves to resemble a harmful species (model). The mimic gains protection from predators. However, if mimics become too common, predators learn that the pattern is not reliably associated with danger, and the advantage of mimicry is lost. Therefore, mimics are favored when they are rare relative to the model.
- **Host-parasite interactions:** A rare parasite genotype may be able to infect hosts that have not evolved resistance to it. Once it becomes common, hosts evolve resistance, and its advantage disappears. This is often called the “Red Queen” effect.

(c) Summary of Frequency Dependent Selection Types

Type	Effect on Rare Alleles	Examples
Positive frequency dependence	Rare alleles are at a disadvantage; common alleles are favored	Warning coloration in toxic species; resistance alleles in parasites
Negative frequency dependence	Rare alleles have an advantage; common alleles are at a disadvantage	Rare male advantage; Batesian mimicry; host-parasite coevolution

Table 1: Summary of frequency dependent selection types.

Mutation as a Force of Evolution

Mutation is the ultimate source of all genetic variation. Without mutation, evolution would eventually stop because there would be no new alleles to act upon. However, mutation alone is a weak evolutionary force because mutation rates are typically very low.

(a) One-Locus, Two-Allele Model with Mutation

Consider a gene with two alleles: A_1 and A_2 .

Let:

- p = frequency of A_1
- q = frequency of A_2
- μ = mutation rate from A_1 to A_2 per generation
- ν = mutation rate from A_2 to A_1 per generation

Mutation rates are typically very small, on the order of 10^{-5} to 10^{-6} per generation.

(i) Unidirectional Mutation

In unidirectional mutation, mutations occur in only one direction. For example, A_1 mutates to A_2 , but A_2 does not mutate back to A_1 .

Let μ = mutation rate from A_1 to A_2 . $\nu = 0$.

Change in allele frequency due to mutation alone:

- Each generation, a proportion μ of A_1 alleles become A_2
- Therefore, the frequency of A_1 in the next generation is:

$$p(t+1) = p(t) - \mu p(t) = p(t) \times (1 - \mu)$$

- The change in allele frequency is:

$$\Delta p = p(t+1) - p(t) = -\mu p(t)$$

Frequency after t generations: Starting from an initial frequency p_0 , after one generation:

$$p_1 = p_0 \times (1 - \mu)$$

After two generations:

$$p_2 = p_1 \times (1 - \mu) = p_0 \times (1 - \mu)^2$$

Therefore, after t generations:

$$p_t = p_0 \times (1 - \mu)^t$$

Equilibrium:

- Without other forces, A_1 will eventually be lost ($p \rightarrow 0$) and A_2 will become fixed ($q \rightarrow 1$)
- The rate of loss is very slow because μ is very small
- For example, if $\mu = 10^{-5}$, after 10,000 generations, $p_t = p_0 \times (1 - 10^{-5})^{10000} \approx p_0 \times 0.905$

(ii) Bidirectional Mutation

In bidirectional mutation, mutations occur in both directions.

Let:

- μ = mutation rate from A_1 to A_2
- ν = mutation rate from A_2 to A_1

Change in allele frequency due to mutation alone:

- A_1 is lost by mutation to A_2 at rate $\mu \times p$
- A_1 is gained by mutation from A_2 at rate $\nu \times q$
- Therefore:

$$\Delta p = -\mu p + \nu q$$

Equilibrium: At equilibrium, $\Delta p = 0$:

$$\begin{aligned} -\mu p + \nu q &= 0 \\ -\mu p + \nu(1 - p) &= 0 \\ -\mu p + \nu - \nu p &= 0 \\ \nu &= p(\mu + \nu) \end{aligned}$$

Therefore:

$$\hat{p} = \frac{\nu}{\mu + \nu}$$

Similarly:

$$\hat{q} = \frac{\mu}{\mu + \nu}$$

Key point: Even without selection, bidirectional mutation maintains both alleles at equilibrium frequencies determined by the mutation rates. However, because mutation rates are very small, the equilibrium frequency of the allele that arises by mutation is also very small unless the mutation rates are nearly equal.

(b) Mutation-Selection Balance

In nature, mutation and selection often act simultaneously. Mutation introduces deleterious alleles, while selection removes them. An equilibrium is reached where the rate of introduction of deleterious alleles by mutation equals the rate of removal by selection.

Consider a deleterious allele A_2 that is maintained by recurrent mutation from A_1 to A_2 . Let:

- μ = mutation rate from A_1 to A_2 per generation
- s = selection coefficient against the deleterious allele

The equilibrium frequency of the deleterious allele ($\hat{q}$) depends on whether it is dominant or recessive.

(i) Deleterious Allele is Dominant

If the deleterious allele A_2 is dominant, heterozygotes (A_1A_2) and homozygotes (A_2A_2) both have reduced fitness. Selection acts against A_2 even when it is in heterozygotes.

The equilibrium frequency is approximately:

$$\hat{q} \approx \frac{\mu}{s}$$

Example: If $\mu = 10^{-5}$ and $s = 0.1$, then:

$$\hat{q} = \frac{10^{-5}}{0.1} = 10^{-4} \text{ (0.01\% or 1 in 10,000)}$$

Key point: Because selection is strong against dominant deleterious alleles (acting immediately in heterozygotes), the equilibrium frequency is very low.

(ii) Deleterious Allele is Recessive

If the deleterious allele A_2 is recessive, only A_2A_2 homozygotes have reduced fitness. Heterozygotes (A_1A_2) are unaffected and can “hide” the deleterious allele.

The equilibrium frequency is approximately:

$$\hat{q} \approx \sqrt{\frac{\mu}{s}}$$

Example: If $\mu = 10^{-5}$ and $s = 0.1$, then:

$$\hat{q} = \sqrt{\frac{10^{-5}}{0.1}} = \sqrt{10^{-4}} = 0.01 \text{ (1\% or 1 in 100)}$$

Key point: Because selection cannot “see” the recessive deleterious allele in heterozygotes, it can accumulate to much higher frequencies than a dominant deleterious allele. This is why many recessive genetic disorders (e.g., cystic fibrosis, sickle cell anemia) have relatively high carrier frequencies even though the disease is severe.

(c) Comparison of Equilibrium Frequencies

Type	Equilibrium Frequency ($\hat{q}$)	Example with $\mu = 10^{-5}$, $s = 0.1$
Dominant deleterious	$\frac{\mu}{s}$	10^{-4} (0.01%)
Recessive deleterious	$\sqrt{\frac{\mu}{s}}$	10^{-2} (1%)

Table 2: Comparison of equilibrium frequencies for dominant and recessive deleterious alleles.

The recessive deleterious allele can be **100 times more common** than a dominant deleterious allele with the same mutation rate and selection coefficient.

(d) A Note on Eugenics

Eugenics is the idea of improving the human population by selective breeding – encouraging individuals with “desirable” traits to reproduce and preventing those with “undesirable” traits from reproducing.

Why eugenics is problematic:

1. **Ethical violations:** Eugenics involves coercive control over reproduction, violating fundamental human rights and bodily autonomy. Historically, eugenics programs have led to forced sterilizations, discrimination, and atrocities.
2. **Scientific misunderstanding:** Even for simple Mendelian disorders, the mutation-selection balance shows that eliminating a recessive deleterious allele by selection alone is practically impossible. Deleterious alleles are constantly reintroduced by mutation and can persist in heterozygotes where they are “hidden” from selection. Attempting to eliminate them would require many generations of selective breeding and would never achieve complete elimination.
3. **Definition of “desirable” is subjective:** What is considered a “desirable” trait is culturally determined and changes over time. There is no objective scientific basis for labeling certain traits as superior or inferior.
4. **Genetic diversity is valuable:** Reducing genetic diversity makes populations more vulnerable to diseases and environmental changes. Many traits considered “disorders” today (e.g., sickle cell trait) actually confer advantages in certain environments (e.g., resistance to malaria).
5. **Complex traits are not simply inherited:** Most human traits (intelligence, behavior, health) are influenced by many genes interacting with environmental factors. They cannot be “bred out” or “bred in” through simple selection.

Modern genetics and medicine focus on treating genetic disorders, providing reproductive choices (e.g., genetic counseling, prenatal testing) without coercion, and respecting individual autonomy. This is fundamentally different from eugenics.

(e) Summary Table: Mutation and Mutation-Selection Balance

Scenario	Key Equation	Equilibrium Frequency
Unidirectional mutation	$p(t+1) = p(t) \times (1 - \mu)$	$p_t = p_0 \times (1 - \mu)^t$
Bidirectional mutation	$\Delta p = -\mu p + \nu q$	$\hat{p} = \nu / (\mu + \nu)$
Dominant deleterious	Selection removes heterozygotes	$\hat{q} \approx \mu / s$
Recessive deleterious	Selection removes only homozygotes	$\hat{q} \approx \sqrt{\mu / s}$

Table 3: Summary of mutation and mutation-selection balance models.